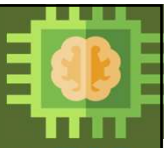
Elective Course
Course Code: CS4103
Autumn 2026-27



Lecture #10

Artificial Intelligence for Data Science

Module-2: PROBLEM SOLVING BY SEARCH

Introduction to Informed Search [Part-II]
(Greedy Best-First Search, A* Search)

Course Instructor:

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Department of Computational and Data Sciences
Indian Institute of Science Education and Research Kolkata, India 741246

Informed Search Strategies



- **Main Idea**
  - Use the knowledge of the problem domain to build an evaluation function f
  - For, every node n in the search space, $f(n)$ quantifies the desirability of expanding n in order to reach the goal
  - Then use the desirability value of the nodes in the fringe to decide which node to expand next
  - The evaluation function f is typically an imperfect measure of the goodness of the node
  - i.e., the right choice of nodes is not always the one suggested by f

Standard Assumptions on Search Spaces



- The cost of a node increases with the node's depth
- Transitions costs are non-negative and bounded below, i.e., there is a $\epsilon > 0$ such that the cost of each transition is $\geq \epsilon$
- Each node has only finitely-many successors

Best-First Search



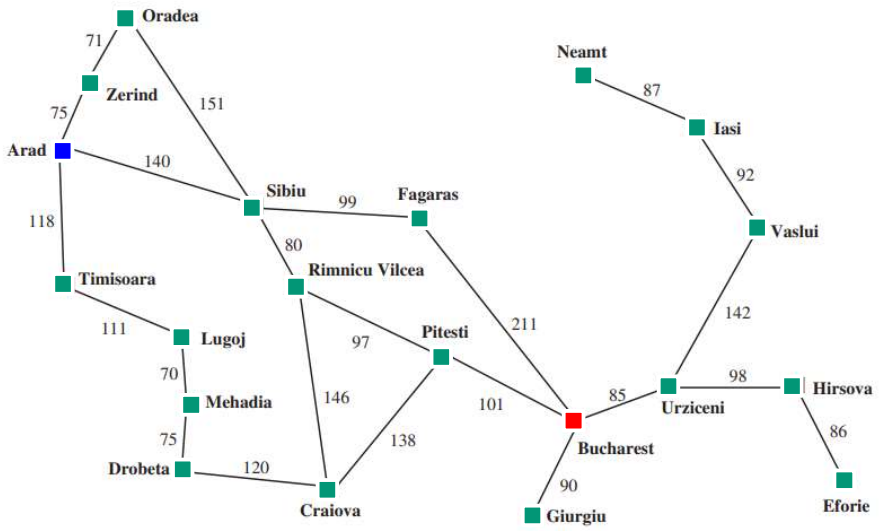
- **Idea:** use an evaluation function estimating the desirability of each node
- **Strategy:** Always expand the most desirable unexpanded node
- **Implementation:** the fringe is a priority queue sorted in decreasing order of desirability
- **Special cases:**
  - greedy best-first search
  - A* search
- Best-first is a family of search strategies, each with a different evaluation function
- Typically, use estimates of the cost of reaching the goal and try to minimize it

Greedy Best-First Search

- Evaluation function $f(n) = h(n)$ (heuristics)
  = estimate cost of cheapest path from node n to closest goal
- E.g., $h_{SLD}(n)$ = straight-line distance from a city n to the destination city.
- **Greedy search expands the node that appears to be closest to goal**

Romania with Step Costs in Km

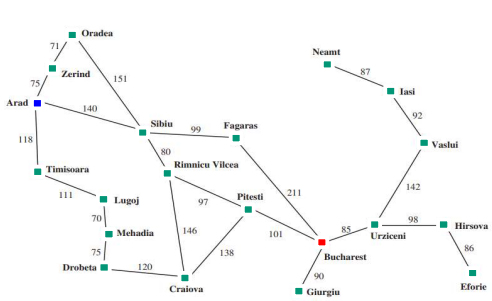
Currently in Arad. Find a route to drive to Bucharest.



Straight-line distance to Bucharest	
Arad	366
Bucharest	0
Craiova	160
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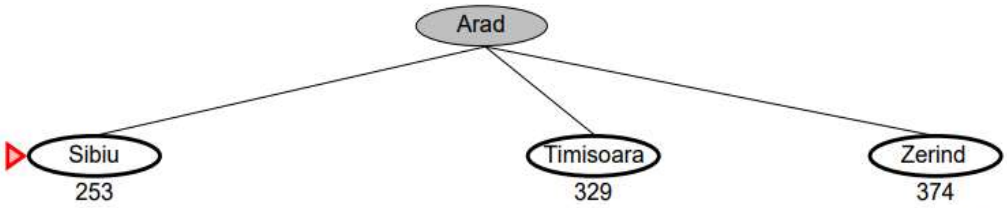
Greedy Search Example

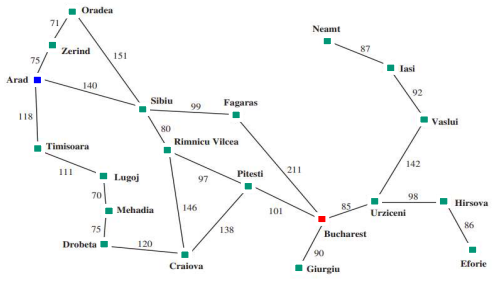




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Greedy Search Example

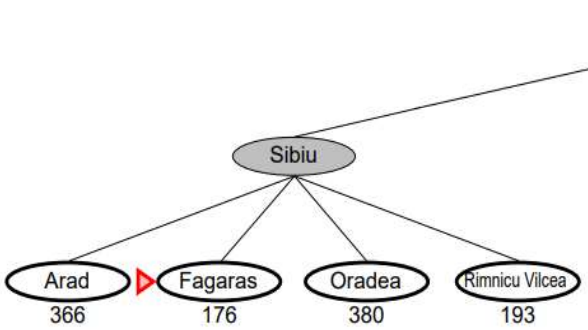


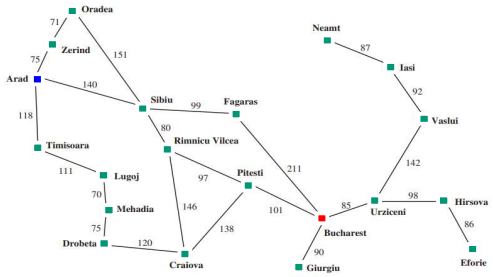


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Greedy Search Example





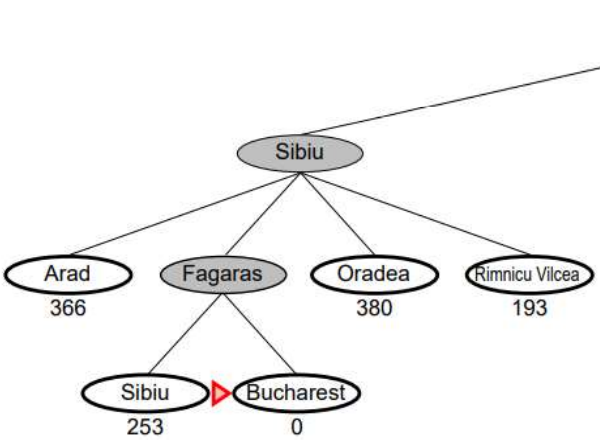


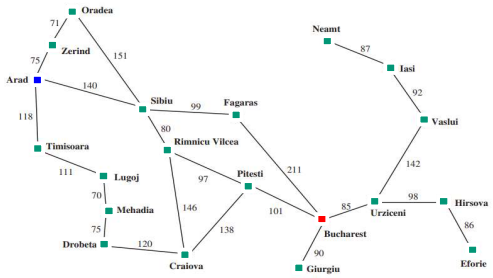
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Greedy Search Example







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Properties of Greedy Best-First Search

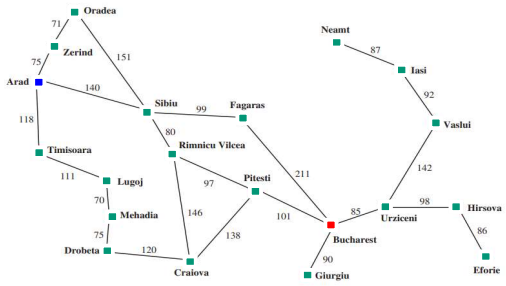


Complete? Only in finite spaces with repeated-state checking
Otherwise, can get stuck in loops:
Iasi $\rightarrow$ Neamt $\rightarrow$ Iasi $\rightarrow$ Neamt $\rightarrow$

Time complexity? $O(b^m)$ — may have to expand all nodes

Space complexity? $O(b^m)$ — may have to keep most nodes in memory

Optimal? No



A* — A Better Best-First Strategy



Greedy Best-first search

- minimizes estimated cost $h(n)$ from current node n to goal
- is informed but almost always suboptimal and incomplete

Uniform cost search

- minimizes actual cost $g(n)$ to current node n
- is, in most cases, optimal and complete but uninformed

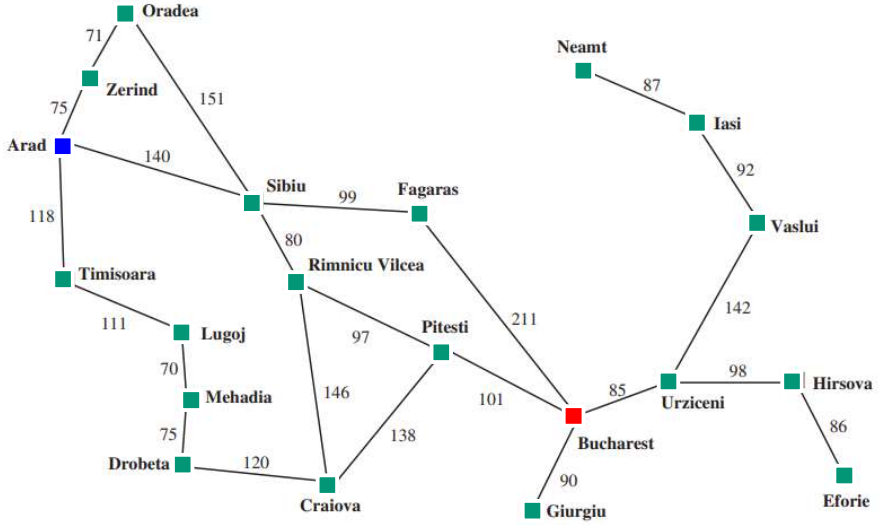
A* search

- combines the two by minimizing $f(n) = g(n) + h(n)$
- is, *under reasonable assumptions*, optimal and complete, and also informed

Romania with Step Costs in Km

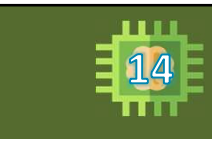


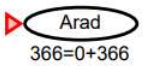
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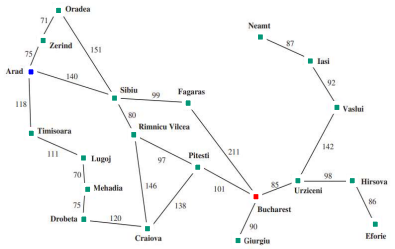
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A* Search Example

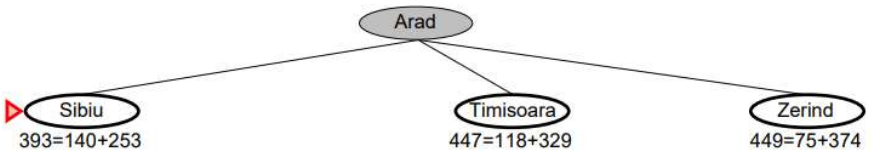




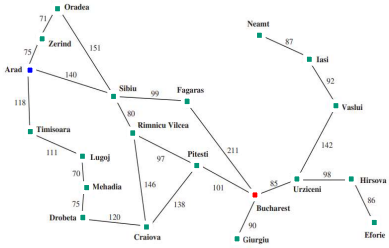
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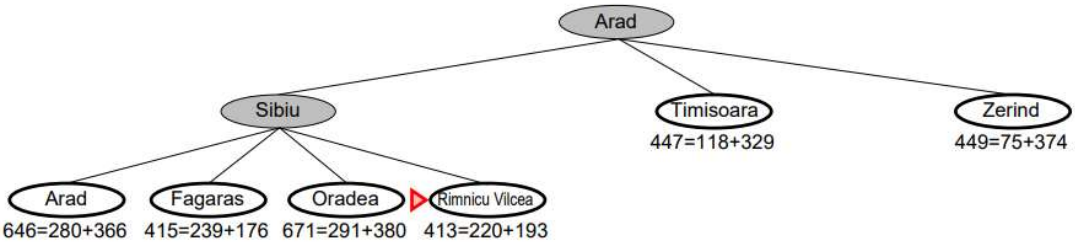
A* Search Example



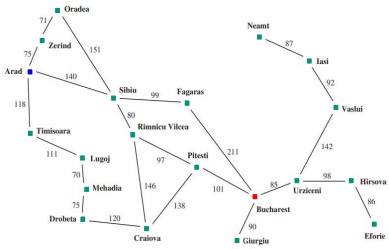
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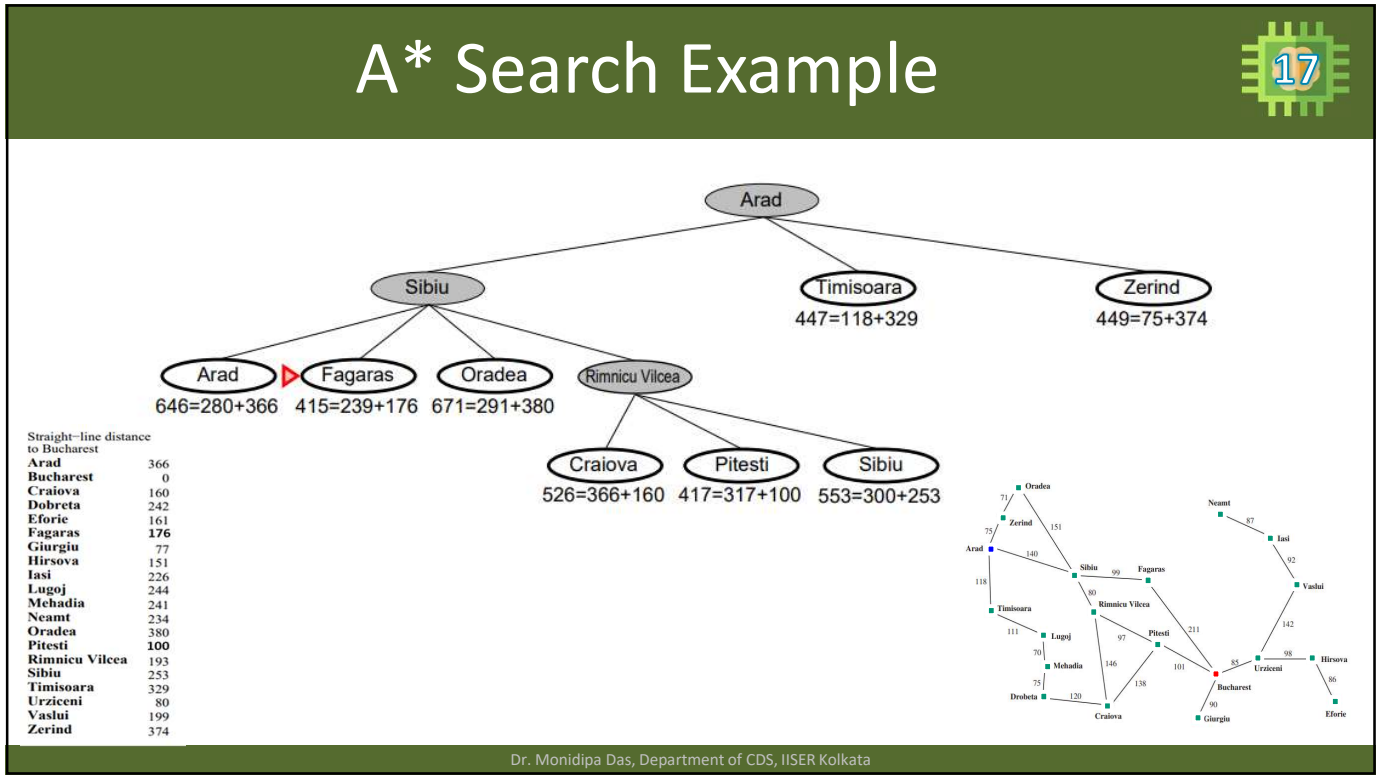
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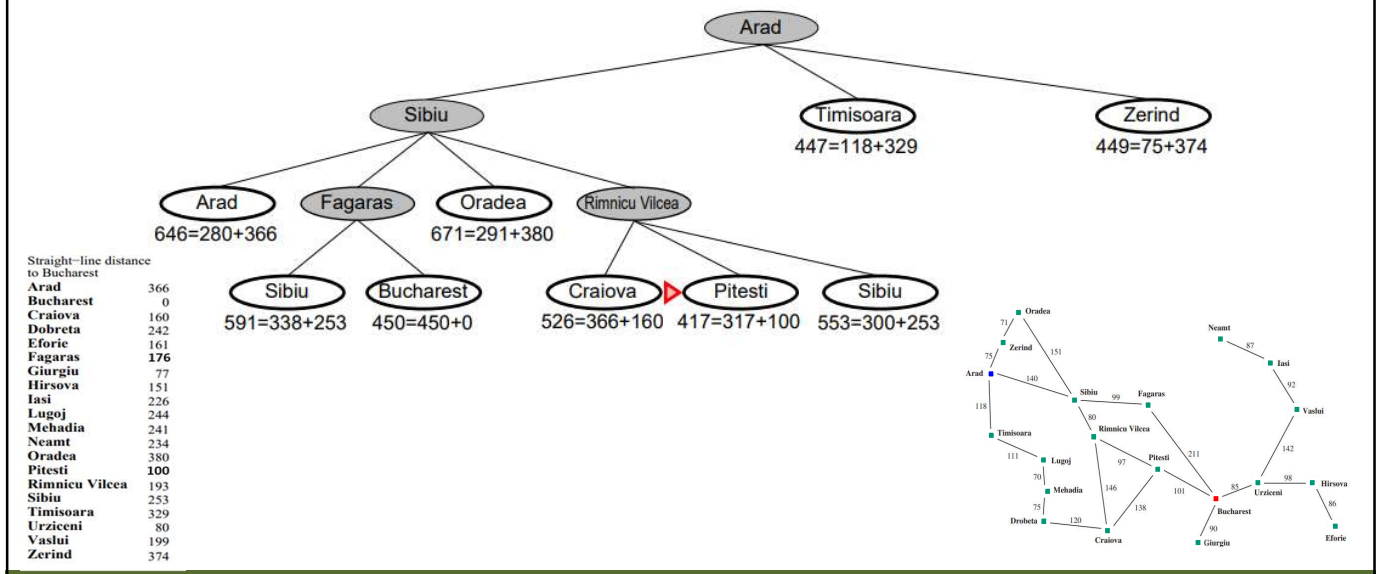
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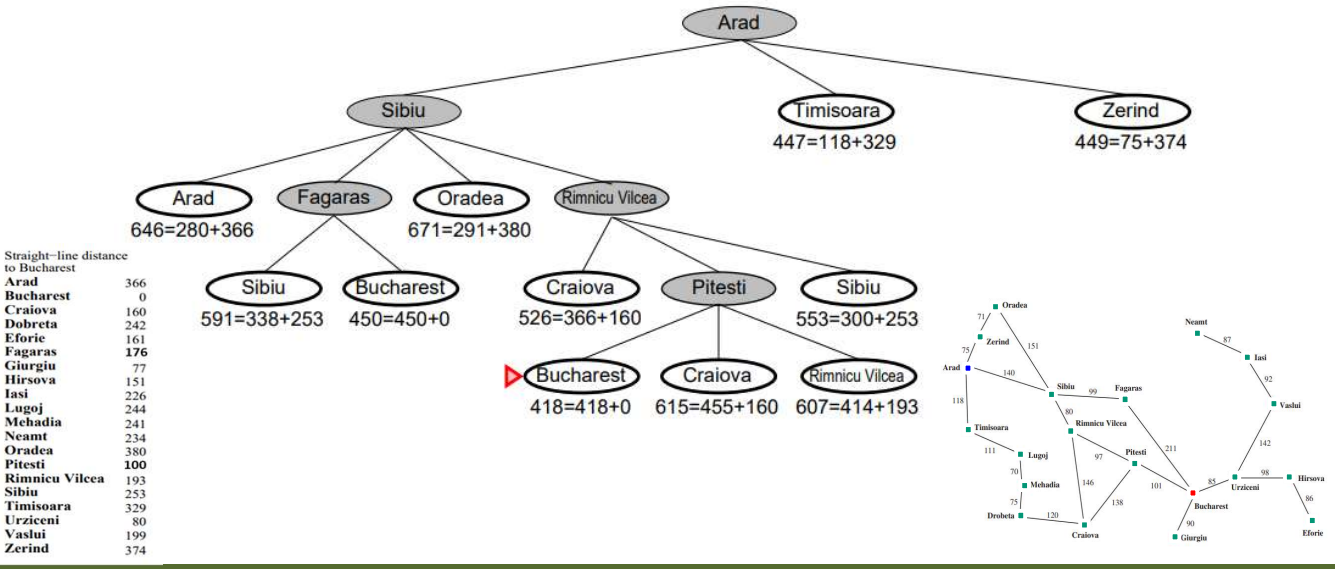
A* Search Example



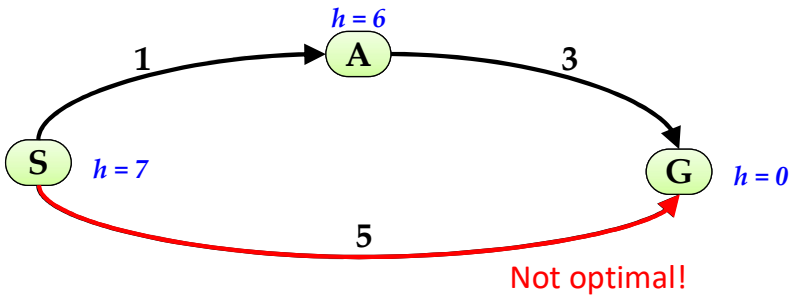
A* Search Example

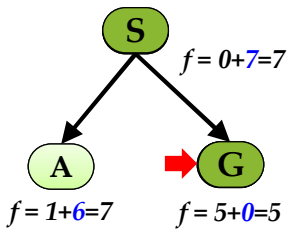


A* Search Example



Is A* Optimal?





What went wrong?

- Actual cost to reach the goal < estimated cost of reaching the goal
- We need estimates to be less than actual costs!

A* with TREE-SEARCH is optimal if $h(n)$ is an **admissible heuristic**

Admissible Heuristics



- A heuristic $h(n)$ is *admissible* **if it never overestimates the actual/true cost to reach the goal**:

  $$0 \leq h(n) \leq h^*(n)$$

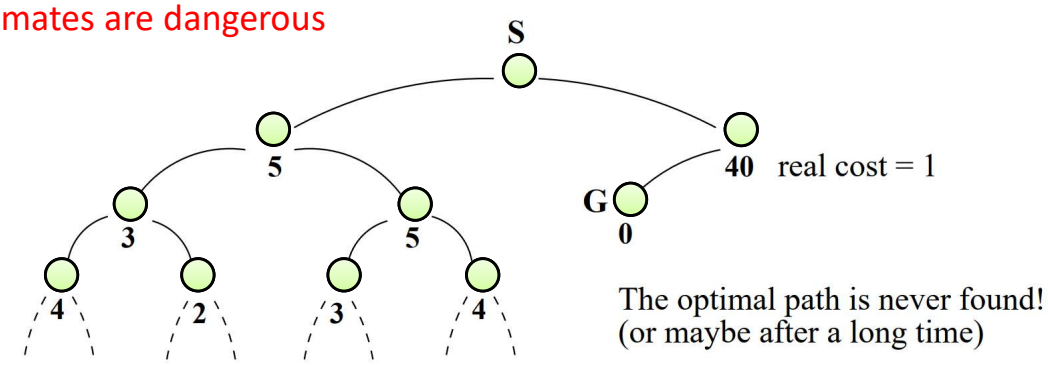
  where $h^*(n)$ is the true cost to a nearest goal

- Admissible heuristics are by nature optimistic
- E.g., $h_{SLD}(n)$ never overestimates the actual road distance
- Finding good, cheap, admissible heuristics is the key to success

A* Search: Why an Admissible Heuristic



- If h is admissible, $f(n)$ never overestimates the actual cost of the best solution through n
- Overestimates are dangerous



Optimality of A* Tree Search

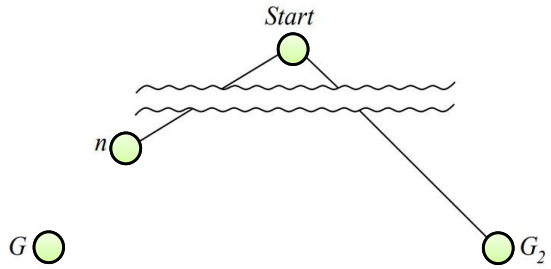


- Suppose some suboptimal goal G_2 has been generated and is in the queue. Let n be an unexpanded node on a least-cost path to an optimal goal G

- $f(G_2) = g(G_2)$ since $h(G_2) = 0$
  $> C^*$ since G_2 is suboptimal
  $\geq g(n) + h(n)$ since ***h* is admissible**

- So, $f(G_2) \geq f(n)$

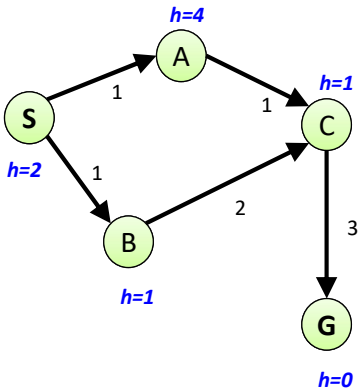
- Since $f(G_2) \geq f(n)$,
  A* will never select G_2 for expansion



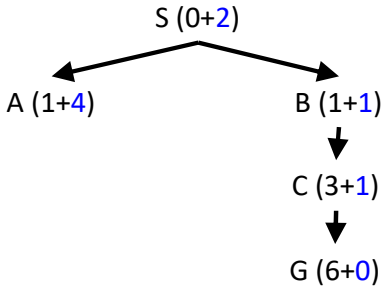
Optimality of A* Graph Search



State space graph



Search tree



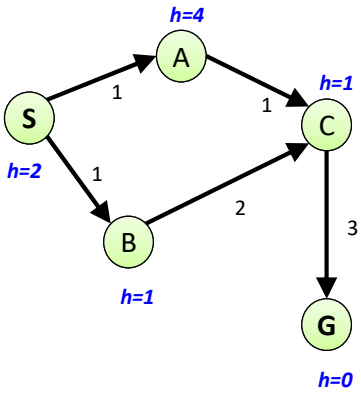
A* Graph Search Gone Wrong?

Simple check against expanded set blocks C

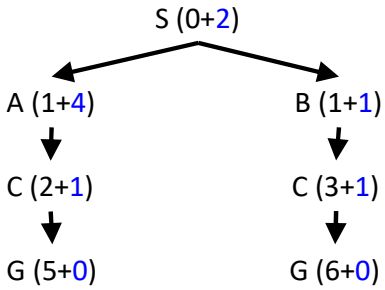
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State space graph



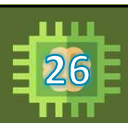
Search tree



The problem can be fixed by imposing requirement of **consistency** on h

A* Graph Search Gone Wrong?

Simple check against expanded set blocks C
Fancy check allows new C if cheaper than old but requires recalculating C's descendants



Questions?